

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: INTERNET PROGRAMMING

COURSE CODE: CSA4305

COURSE OUTCOME COVERED

CO5: Construct interoperable web services by leveraging JAX-RPC, WSDL, SOAP, and XML Schema for seamless data exchange across distributed systems.(L4,L5)

Assessment Tool : Alternative Solution Design

Weightage: 20%

QUESTIONS:

Total Marks: 25

1. A company needs to develop a web service for a mobile and web application. Analyze both **REST and SOAP** approaches and design an appropriate solution by selecting one. Justify your choice based on performance, scalability, and security requirements.
2. An online platform expects a large number of users accessing its services simultaneously. Design a **scalable service architecture** showing how client, server, API, and database interact. Propose an alternative design to improve scalability and justify your approach.
3. A banking application requires secure communication between client and server. Analyze available communication protocols and **select the most suitable protocol**. Justify your selection based on security, reliability, and performance.
4. An application needs to integrate with third-party services such as payment gateways or maps. Design an **API integration strategy**, including request handling, authentication, and response processing. Suggest an alternative approach to improve efficiency or reliability.
5. A web service frequently encounters errors during client requests. Design an **error handling and fault tolerance mechanism**. Propose alternative strategies to improve system reliability and ensure smooth user experience.

Code of Conduct:

I P Madhavi/192372133 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Madhavi

RUBRICS FOR CALCULATION:

Criteria	Weight age (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly defines problem, objectives, and all requirements accurately	Understands problem with minor gaps	Partial understanding; misses key requirements	Misinterprets or fails to understand problem
Generation of Solutions	25%	Develops multiple innovative and feasible alternatives with clear differentiation	Generates relevant alternatives with minor limitations	Limited or repetitive alternatives	Fails to generate meaningful alternatives
Evaluation of Alternatives	25%	Critically compares alternatives using appropriate criteria and metrics	Compares alternatives with some justification	Basic or superficial comparison	No proper comparison conducted
Solution Selection	20%	Selects best solution with strong justification and decision rationale	Selects suitable solution with moderate justification	Selection is weakly justified	No clear or justified selection
Feasibility	10%	Applies relevant concepts ensuring high feasibility and practicality	Applies concepts with minor issues	Limited feasibility or incorrect application	Solution is impractical or conceptually incorrect

1. REST vs SOAP – Analyze and Select One (5 Marks)

Introduction

Web services allow communication between applications over the internet. The two common approaches are REST (Representational State Transfer) and SOAP (Simple Object Access Protocol).

REST vs SOAP Comparison

Feature	REST	SOAP
Architecture	Architectural style	XML-based protocol
Data Format	JSON, XML, HTML	XML only
Performance	Fast and lightweight	Slower due to XML
Scalability	Highly scalable	Less scalable
Security	HTTPS, OAuth, JWT	WS-Security support
Best Use	Mobile and web applications	Enterprise applications

Selected Approach: REST

REST is chosen because the company needs both mobile and web applications.

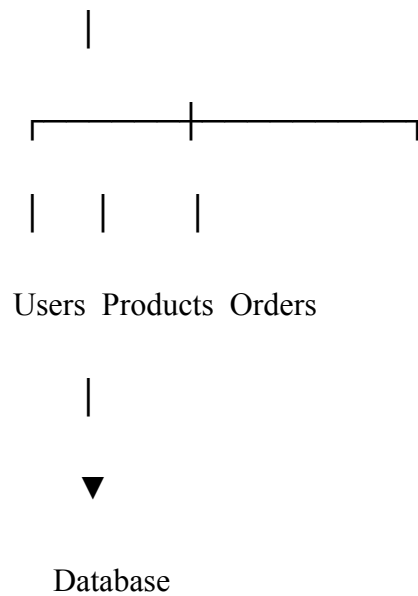
REST Architecture Flow Chart

Mobile App / Web App

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REST API (HTTP)



Why REST is Selected?

Requirement Justification

Performance JSON is lightweight and transfers data quickly.

Scalability Stateless architecture supports millions of requests.

Security Uses HTTPS, OAuth, and JWT authentication.

Flexibility Easy integration with Android, iOS, and web applications.

Advantages of REST

- Fast response using JSON.
- Easy to develop and maintain.
- Platform independent.
- Supports cloud-based applications.

Conclusion: REST is the best choice for modern mobile and web applications because it provides high performance, scalability, and secure communication.

2. Scalable Service Architecture (5 Marks)

Introduction

A scalable architecture allows many users to access the application simultaneously without reducing performance.

Basic Scalable Architecture

Flow Chart

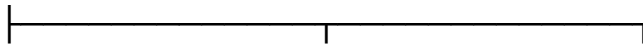
Users



Load Balancer



API Gateway



Application Authentication Notification

Server

Service

Service

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Database Server

Components of Architecture

Component	Function
Client	Sends API requests from mobile/web app.
Load Balancer	Distributes traffic among multiple servers.
API Gateway	Handles routing, authentication, and API management.
Application Server	Processes business logic.
Database	Stores application data.

Alternative Design (Microservices)

Flow Chart

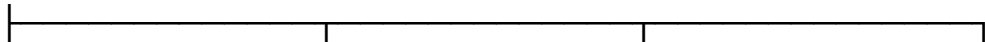
Users

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API Gateway

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User Service Product Service Payment Service Order Service



Separate Databases

Why Microservices Improve Scalability?

Feature

Benefit

Independent Services Each service scales separately.

Fault Isolation Failure in one service doesn't stop others.

Easy Deployment Services can be updated independently.

Conclusion

Microservices with an API Gateway provide better scalability and reliability for applications with a large number of users.

3. Secure Communication Protocol for Banking Application (5 Marks)

Introduction

Banking applications require secure communication to protect user credentials and financial transactions.

Protocol Comparison

Protocol	Security	Performance
HTTP	No encryption.	Fast but insecure.
HTTPS	SSL/TLS encryption.	Fast and secure.
SOAP + WS-Security	Message-level security.	Reliable but slower.
REST + HTTPS	Token-based security.	Lightweight and scalable.

Selected Protocol: HTTPS with TLS

Secure Communication Flow Chart

Banking Mobile App

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| HTTPS Request



SSL/TLS Encryption

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Bank Server

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Encrypted Database Access

Why HTTPS?

Requirement	Reason
Security	Encrypts data using SSL/TLS.
Reliability	Protects against tampering and man-in-the-middle attacks.
Authentication	Supports certificates and secure login tokens.
Performance	Provides secure communication with minimal overhead.

Additional Security Measures

- JWT Authentication.
- OAuth 2.0.
- Multi-Factor Authentication (MFA).
- Session timeout and encryption.

Conclusion

HTTPS with TLS is the most suitable protocol because it offers strong encryption, authentication, reliability, and good performance.

4. API Integration Strategy with Third-Party Services (5 Marks)

Introduction

Applications often integrate with third-party services like payment gateways, maps, SMS, or email APIs.

API Integration Flow Chart

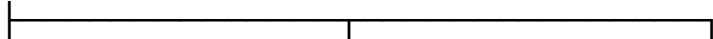
User



Application



API Gateway



Payment API Maps API Email API



Third-Party Service Response



Application Response to User

Integration Steps

Step	Description
1. Send Request	Application sends API request with required parameters.

2. Authentication API key, OAuth token, or JWT is attached.
3. Process Request Third-party server processes the request.
4. Receive Response Application receives JSON/XML response.
5. Display Result Processed response is shown to the user.

Authentication Methods

Method	Purpose
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API Key	Identifies authorized application.
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OAuth 2.0	Secure user authorization.
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JWT Token	Secure authentication for APIs.
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Alternative Strategy

API Gateway with Caching

Client

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API Gateway

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└─ Cache

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| Cached Response

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Third-Party API

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Application

Benefits

- Faster responses through caching.
- Reduced API calls.
- Improved reliability during high traffic.

Conclusion

API Gateway with caching and OAuth authentication improves efficiency, security, and reliability when integrating third-party services.

5. Error Handling and Fault Tolerance Mechanism (5 Marks)

Introduction

Error handling ensures that applications respond correctly when failures occur and continue providing services to users.

Error Handling Flow Chart

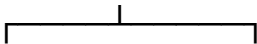
Client Request



API Server



Validate Request



Valid Invalid



Process Error Response



Database





Success Response

Common Web Service Errors

Error	Description
400 Bad Request	Invalid client request.
401 Unauthorized	Authentication failed.
404 Not Found	Requested resource unavailable.
500 Internal Server Error	Server processing failure.
503 Service Unavailable	Server temporarily unavailable.

Fault Tolerance Mechanism

Flow Chart

Client Request

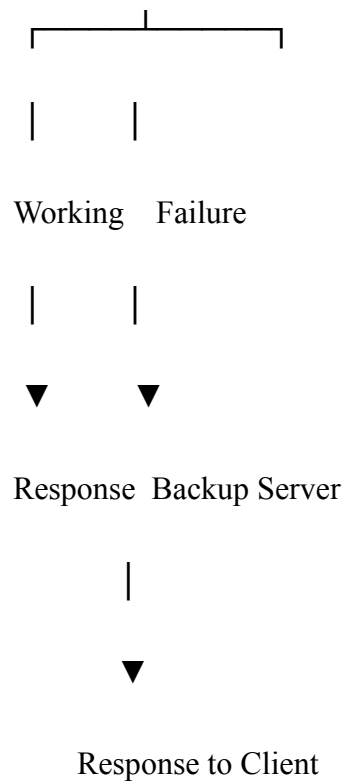


API Gateway



Primary Server





Alternative Reliability Strategies

Strategy	Benefit
Retry Mechanism	Automatically retries failed requests.
Circuit Breaker	Stops repeated requests to failed services.
Load Balancing	Redirects requests to healthy servers.
Caching	Provides previously stored responses.
Logging and Monitoring	Detects and fixes issues quickly.

Benefits of Fault Tolerance

- High system availability.
- Better user experience.

- Reduced downtime.
- Faster recovery from failures.